

Facilities and Other Resources

Phase I

Using AI to assist high fidelity multiphase flow modeling in making atomization and spray predictions.

PI: Mark Sussman, Department of Mathematics, Florida State University

CO-PI: Kourosh Shoele, FAMU/FSU Mechanical Engineering

CO-PI: Marco Arienti, Sandia National Labs, Livermore

Laboratory Resources: The FAMU-FSU College of Engineering performance site facilities used in this project are located within the innovation park at the joint college of engineering in Tallahassee. CO-PI Shoele is a member of Florida Center for Advanced Aero-Propulsion (FCAAP) and has a 270 square foot lab equipped with computers. Office space for graduate students are provided by FCAAP.

Computer (Shoele): CO-PI Shoele's laboratory is equipped with over a dozen PC workstations. Key software includes Tecplot, GridGen, Intel compilers. The department of Mechanical Engineering and FCAAP provide the software as well as printing facilities.

Computer (Sussman): PI Sussman has access in the math department to 8 4-core computers, 1 32 core machine, and 1 8-core machine. The 8-core machine has an Nvidia GPU card in it.

Computing Facilities: There are a number of major computing facilities at FSU-FAMU that are available to the PIs Sussman, Shoele, and students in FSU-FAMU for this project. 1) HPC Cluster: 647 node, 10772 core, cluster with CUDA GPU capability attached to 256 TB Panasas file system that is mounted on all of the HPC compute nodes over a dedicated 1Gbps network. 2) Spear cluster: 88 x86 64-bit process cores linked together by a QDR Infiniband network. Eight of the nodes also connect to a PCI chassis equipped with 16 NVIDIA Tesla gpGPU cards (m2050) for a total 7,168 GPU cores for a peak system performance of 19.8 TFLOPS. 3) CAPS Real Time Digital Simulator (RTDS™): A parallel-processing machine (performance at over 110 GigaFLOPS) with "14- rack" system consisting of ~380 parallel processors and employing optimized EMTP-based nodal network simulation methods.

